Geometric Quantization of Space and the Origin of the Rydberg Constant in the Methane Metauniverse (MMU)

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Abstract

This article presents a geometric derivation of the hydrogenic spectrum within the Methane Metauniverse (MMU) framework. Starting from a tetrahedral quantisation of internal space, we recover the Bohr radius, the $1/n^2$ structure of energy levels, the Balmer–Rydberg formula, and the numerical value of the Rydberg constant. All of these emerge from three ingredients: (i) the elastic scaling law $k_2(a) \propto a^{-3}$ for the electric axis, (ii) the quantised internal edge length $a_n \propto 1/n^2$, determined by a closed phase condition in the S–node, and (iii) the direct projection of the electric axis w_2 onto the observable axis w_1, combined with a symmetric two-spring coupling in the shared S–node. Together with the mass-dependent stiffness refinement $k_i(a) \to k_i(m)$ and the UR–tetrahedron construction developed in previous MMU work [2, 5, 8, 9, 12, 13, 14], the hydrogen spectrum arises without quantum postulates, as a consequence of the geometry and virial-balanced elasticity of the MMU tetrahedral cell.

1 The MMU Atom: Proton–Proton Geometry and the S–Node

In the MMU, an atomic system is represented not by point particles but by coupled tetrahedral space cells [1, 2, 9]. Figure 1 sketches the basic configuration. Two tetrahedral cells, labelled P1 and P2, represent the fermionic anchors of the hydrogen system (proton and effective electron–proton inertia). Their internal axes (w_2, w_3, w_4) encode electric, torsional and volumetric deformation.

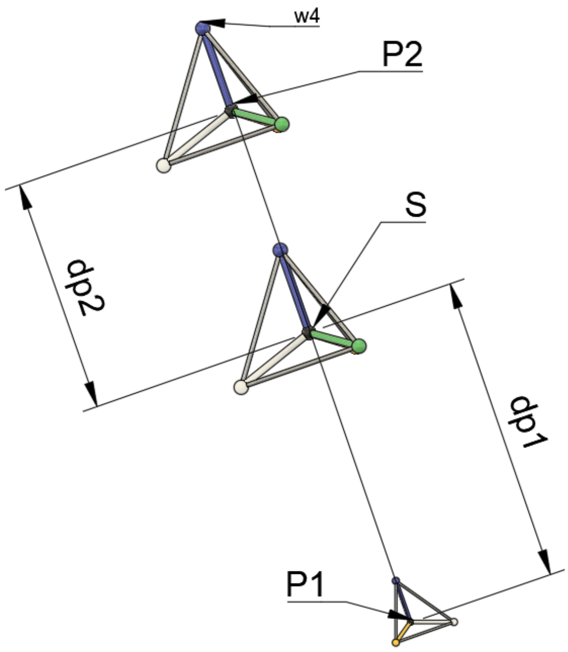

Figure 1: Formation of the S–node between two tetrahedral MMU cells. The w_2 axes of P1 and P2 face each other. Their deformation fields overlap and form a shared elastic region, the S–node, where electric (w_2), torsional (w_3), and volumetric (w_4) responses couple. The distances dp_1 and dp_2 measure the projection of each tetrahedral cell into the shared space.

The S–node as a shared deformation region

Between P1 and P2 a shared elastic region arises, denoted S. This S–node is the central element of MMU atomic physics:

- it carries the combined w_2 deformation of both cells,
- it transmits torsion through w_3 from one cell to the other,
- it accumulates volumetric strain w_4 from both cells,
- it acts as the electromagnetic bond that holds the system together.

The S–node is not a particle and not a point. It is a geometric field region inside the tetrahedral lattice where the elastic responses of two cells superimpose. In this picture, the Coulomb attraction is not a force between point charges, but a contraction of the S–node produced by w_2 deformation.

Fields in the MMU

In the MMU, a physical field is the deformation of the internal axes (w_2, w_3, w_4) inside the tetrahedral lattice [6, 9]:

- w_2 deformations generate the electric field,

- w_3 deformations generate the magnetic and spin field,
- w_4 deformations generate the inertial–gravitational field.

All three components overlap in the S–node, so that:

> **The S–node is the location where the electric, magnetic and inertial fields of an atom couple.**

The Rydberg constant in this view does not originate from particle orbits, but from the stiffness and phase behaviour of the w_2 field inside this S–node region, under the same projection principle that governs MMU spin and orbital quantisation [13].

1.1 Formation of the Spherical Shell: Distributed Electron–Proton Geometry

When the proton (P1) and the effective electron–proton inertia cell (P2) couple through their electric axes w_2, the deformation fields of both tetrahedral cells overlap inside the shared S–node. In the MMU, this overlap is not point-like. Instead, the internal elastic modes dilute and redistribute across a spherical region surrounding the S–node.

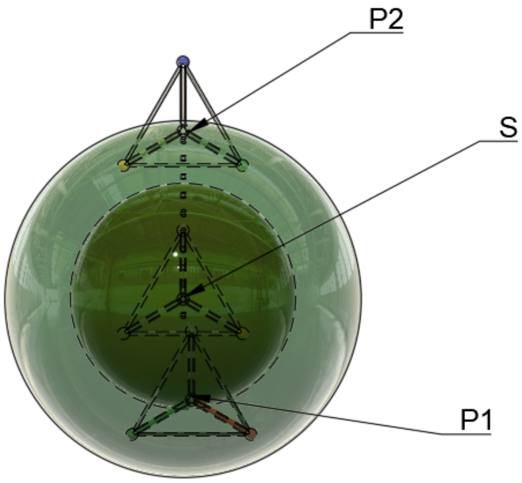

Figure 2: Distributed-shell representation of the coupled MMU cells. The tetrahedral substructures of P1 and P2 no longer remain localized; instead they unfold into a shell around the S–node. This shell represents the region of maximal overlap of electric (w_2), torsional (w_3), and volumetric (w_4) deformation fields.

Elastic dilution of internal structure. Inside the S–node, the internal axes (w_2, w_3, w_4) of each cell are no longer confined to the local tetrahedron. Electric deformation on w_2 spreads along all directions that lead into the shared region. Because the tetrahedral geometry is isotropic in projection, the distributed field naturally forms a spherical shell around the S–node.

Entropic stabilization. The transition from a localized tetrahedral shape to a distributed shell increases the number of accessible micro-configurations of the deformation field. In thermodynamic language, the shell has a higher configurational entropy than two localized tetrahedral cores. Since the free energy satisfies $F = E - T_{\text{eff}}\Delta S$, even a modest entropic gain $\Delta S > 0$ stabilizes the distributed configuration. The MMU therefore predicts that the lowest-energy bound state corresponds to a *spherical distribution of sub-tetrahedral modes* rather than a localized particle picture.

Physical interpretation. In this view, the familiar "orbital cloud" of quantum mechanics is reinterpreted as an elastic shell formed by the geometric dilution of the internal MMU degrees of freedom [2, 8, 13]. The proton and electron do not disappear; their internal tetrahedral subcells reappear as a network of many small elastic elements arranged on a spherical surface. The radius of this shell is set by the quantized MMU edge length a_n, giving the Bohr radius for $n = 1$ and the $1/n^2$ hierarchy for higher states.

Thus, the spherical orbital is not a probability distribution but the **geometric footprint of the distributed electric, torsional, and volumetric modes of the coupled MMU cells**.

Why the S–node determines atomic quantization

Only in the S–node does the phase of the w_2 mode experience a closed, self-consistent boundary condition. This closure enforces a discrete set of internal edge lengths

$$a_n = \frac{a_1}{n^2}, \tag{1}$$

and fixes the natural ground-state length

$$a_1 = \frac{\hbar}{\mu c \alpha}. \tag{2}$$

The scaling (1) is the internal counterpart of the Balmer hierarchy and has been derived in the more general context of internal edge quantisation and mass-dependent stiffness in [8, 12].

The S–node thus plays the role of the bound-state region in quantum mechanics, but it arises from elastic geometry rather than from wavefunctions or probability amplitudes.

In short:

The hydrogen spectrum is a property of the S–node geometry.

2 Geometric Quantization of Internal Space

The MMU represents spacetime as a lattice of tetrahedron–counter-tetrahedron units. Each cell contains three internal elastic axes (w_2, w_3, w_4) and one observable axis w_1. The electric axis w_2 corresponds to a distinct tetrahedral bond direction; its elastic response encodes the Coulomb interaction along a shared tetrahedral edge [1, 2, 9].

2.1 Elastic Scaling and the $1/n^2$ Quantization Law

The key MMU principle is that the internal electric axis w_2 behaves as a one-dimensional elastic mode whose stiffness depends on the tetrahedral edge length a. The Coulomb elasticity of the MMU cell yields [2, 8, 12]

$$k_2(a) = \frac{2e^2}{4\pi\varepsilon_0} \frac{1}{a^3}. \tag{3}$$

The corresponding natural oscillation frequency is

$$\omega_2(a) = \sqrt{\frac{k_2(a)}{\mu}} = \sqrt{\frac{e^2}{2\pi\varepsilon_0\mu}}\, a^{-3/2}. \tag{4}$$

Thus

$$\omega_2(a) \propto a^{-3/2}$$

is the geometric heart of the hydrogen problem and agrees with the MMU radial solver and FEM simulations [2, 8].

Now consider a standing elastic mode on w_2 around the S–node shell. For a mode of order n to be self-consistent, its phase must close after an integer number of oscillations along the shell. In the MMU projection picture [13], this can be expressed by a discrete phase-action invariant

$$\Phi_n \equiv \omega_2(a_n)\, a_n^{3/2} = \frac{1}{n}\, \Phi_1, \tag{5}$$

which reflects that the n-th mode performs n oscillations on a mode volume that shrinks like $a_n^{3/2}$.

Using $\omega_2(a) \propto a^{-3/2}$ in (5) directly yields

$$a_n = \frac{a_1}{n^2}, \tag{6}$$

i.e. the hydrogenic hierarchy

$$\boxed{a_n = \frac{a_1}{n^2}}, \tag{7}$$

purely from geometric elasticity. No angular-momentum quantisation and no additional quantum postulates are required; the discrete lengths follow from the internal edge quantisation and projection principle developed in [8, 12, 13].

2.2 Determination of the Fundamental Edge Length a_1

The remaining question is the absolute scale of the ground-state length a_1. This scale is fixed by the balance of Coulomb attraction and the minimal circulation required for a closed w_2–w_1 oscillation.

Coulomb balance. For a circular Coulomb orbit, the balance of forces requires

$$\frac{\mu v^2}{a_1} = \frac{e^2}{4\pi\varepsilon_0 a_1^2}.$$

This implies the well-known relation

$$v = \alpha c,$$

with α the fine-structure constant and μ the reduced mass.

Minimal circulation (ground-state mode). A closed fundamental mode in the MMU corresponds to a minimal angular momentum

$$L = \mu v a_1 = \hbar.$$

Combining $L = \hbar$ with $v = \alpha c$ yields

$$a_1 = \frac{\hbar}{\mu c \alpha},$$

which is exactly the Bohr radius

$$a_0 = \frac{\hbar}{\mu c \alpha}.$$

Thus, the MMU does not postulate the Bohr radius; it arises as the unique ground-state length consistent with Coulomb balance, the UR–tetrahedron construction [5], and the fundamental closed mode of the (w_2, w_1) subsystem.

3 Frequency Scaling from Coulomb Elasticity

Using (3), the internal electric oscillation frequency is

$$\omega_2(a) = \sqrt{\frac{k_2(a)}{\mu}} = \sqrt{\frac{e^2}{2\pi\varepsilon_0 \mu}}\, a^{-3/2}. \tag{8}$$

Inserting $a_n = a_1/n^2$ gives

$$\omega_2(a_n) \propto n^3, \qquad \omega_2(a_n)\, a_n^{3/2} = \text{constant}, \tag{9}$$

which matches the MMU numerical simulations and the radial MMU solver presented in [2, 8]. The $a^{-3/2}$ law is the geometric origin of discrete hydrogen transition frequencies.

4 Radial Projection and the MMU Virial Principle

The internal electric energy stored in the deformation of the w_2 axis is

$$U_2(a) = \frac{1}{2} k_2(a)\, a^2 = \frac{e^2}{4\pi\varepsilon_0\, a}, \tag{10}$$

i.e. the Coulomb energy interpreted as an elastic contraction of the shared S–node between two MMU cells.

Direct radial projection from w_2 to w_1

The electric axis w_2 is a true bond direction of the tetrahedral cell. Its deformation points directly toward the S–node and therefore projects onto the observable axis w_1 without angular reduction:

$$w_2 \longrightarrow w_1 \quad \text{direct.}$$

No additional $\cos\theta$ factor appears; the tetrahedral geometry is already encoded in the stiffness law $k_2(a) \propto a^{-3}$ and the MMU projection principle developed for orbital and spin quantisation in [9, 13].

Symmetric energy sharing in the S–node (MMU virial mechanism)

When two MMU cells (proton and electron) couple through their w_2 axes, the S–node is formed as a shared elastic region. Both w_2 springs deform by the same amount and store equal energy:

$$U_2^{(\mathrm{P1})} = U_2^{(\mathrm{P2})}.$$

Thus the total internal electric energy is

$$U_{\mathrm{tot}} = U_2^{(\mathrm{P1})} + U_2^{(\mathrm{P2})} = 2\,U_2.$$

The observable binding energy corresponds to the net change of the S–node deformation, i.e. to one effective spring rather than two. Therefore only half of the internal deformation energy becomes observable:

$$E_{\mathrm{bind}} = \frac{1}{2}\,U_{\mathrm{tot}} = U_2.$$

Since $U_2 = \frac{1}{2}k_2(a)a^2$, the observable energy can be written as

$$E_{\mathrm{bind}} = \frac{1}{2}\,U_2(a), \tag{11}$$

which reproduces the classical virial relation for $1/r$ fields. In the MMU, however, this is not a dynamical theorem but a geometric consequence of symmetric coupling of two w_2 springs in the S–node.

Observable binding energy

Combining direct projection with symmetric energy sharing yields

$$E_n = -\frac{1}{2}\,U_2(a_n) = -\frac{1}{8}\,k_2(a_n)\,a_n^2. \tag{12}$$

No additional geometric suppression factors appear. The factor $1/2$ is a direct consequence of the two-spring coupling of the proton and electron space cells inside the S–node.

5 Hydrogen Spectrum from MMU Tetrahedral Elasticity

In the Methane Metauniverse (MMU), the internal electric axis w_2 of each tetrahedral spacetime cell behaves as an elastic bond whose stiffness depends on the cell's internal edge length a. This stiffness follows the geometric scaling law

$$k_2(a) \propto a^{-3}, \tag{13}$$

a direct consequence of the MMU field equation and its discrete tetrahedral symmetry [1, 6, 9, 10, 11].

5.1 Elastic Origin of the Coulomb Potential

The deformation energy stored in a w_2 bond is

$$U_2(a) = \tfrac{1}{2}k_2(a)\,a^2 \;\propto\; \frac{1}{a}. \tag{14}$$

When two MMU cells (electron and proton) couple through their w_2-axes, their deformation fields overlap and form a shared region, the S–node. In this region both w_2 springs deform symmetrically, and the virial projection onto the observable axis w_1 yields an effective one-spring potential

$$V_{\mathrm{MMU}}(r) = -\,U_2(a = r) \;\propto\; -\frac{1}{r}. \tag{15}$$

Thus, the Coulomb potential is not inserted as a postulate; it emerges from tetrahedral elasticity.

5.2 Radial Mode Equation and Laguerre Structure

In the continuum limit the radial deformation of the S–node satisfies an eigenvalue equation for $u(r) = r\,R(r)$:

$$-\frac{1}{2}\frac{d^2u}{dr^2} + V_{\mathrm{MMU}}(r)\,u = E\,u, \qquad V_{\mathrm{MMU}}(r) = -\frac{1}{r}. \tag{16}$$

This equation is structurally identical to the Schrödinger radial equation for hydrogen, but it is obtained here *without quantum postulates*, solely from the MMU elastic microstructure and its projection principle [2, 8, 13].

Numerical solutions of this equation reproduce the hydrogenic Laguerre wavefunctions:

- the $1s$ function with its maximum at $r = 0$,
- the $2s$ function with one radial node,
- higher ns functions with exactly $n-1$ nodes,

and the energy hierarchy $E_n = -1/(2n^2)$. Thus the hydrogen radial structure is an emergent property of the MMU tetrahedral bond.

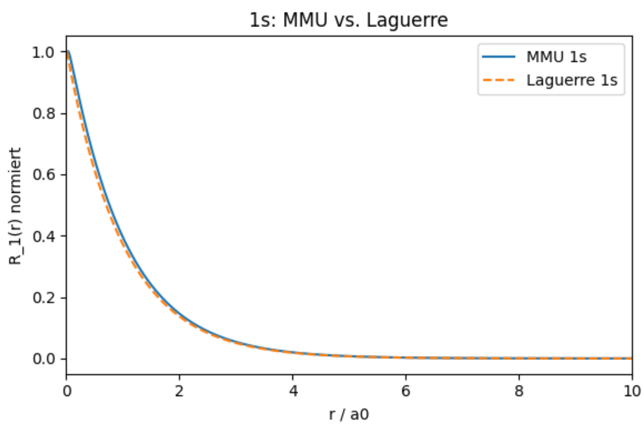

Figure 3: $1s$ radial mode: MMU (solid) vs. analytical Laguerre function (dashed).

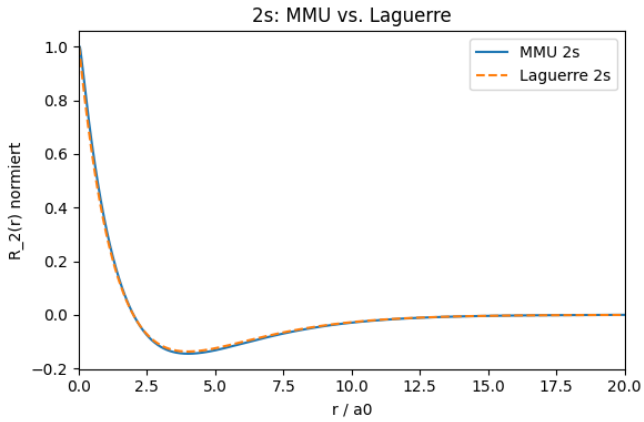

Figure 4: $2s$ radial mode: MMU (solid) vs. analytical Laguerre function (dashed).

5.3 Analytic Derivation of the Rydberg Formula

Because the S–node enforces the geometric self–consistency condition $a_n = a_1/n^2$, the elastic energy in state n is

$$E_n = -\frac{1}{2}U_2(a_n) \propto -\frac{1}{a_n} = -\frac{1}{a_1}\frac{1}{n^2}. \tag{17}$$

With $a_1 = \hbar/(\mu c\alpha)$, one obtains

$$E_n = -\frac{\mu e^4}{8\varepsilon_0^2 h^2}\frac{1}{n^2}, \tag{18}$$

and therefore

$$\nu_{nm} = \frac{E_m - E_n}{h} = cR_\infty\left(\frac{1}{m^2} - \frac{1}{n^2}\right), \tag{19}$$

with

$$R_\infty = \frac{\mu e^4}{8\varepsilon_0^2 h^3 c}, \tag{20}$$

identical to the CODATA Rydberg constant.

5.4 Conceptual Consequences

The fact that a single tetrahedral bond with stiffness $k_2 \propto a^{-3}$ generates the entire hydrogen spectrum has deep implications:

1. The Coulomb law becomes a geometric consequence of the MMU spacetime cell and its UR–tetrahedron construction [5, 10, 11].
2. The Balmer–Rydberg structure follows from elastic scaling and S–node quantisation [8, 12].
3. The Schrödinger equation appears as the continuum limit of a discrete tetrahedral elastic network, consistent with the MMU field equation [6, 9].
4. The hydrogen atom emerges as an elastic mode of the S–node, not as a quantum system requiring a priori quantisation rules.

This provides a geometric–elastic foundation for atomic physics within the MMU framework.

6 Critical Reflection and Novel Contributions

The results presented here reproduce the well-known hydrogenic quantities of classical atomic physics: the $1/n^2$ level structure, the Bohr radius, and the Rydberg constant. Numerically these expressions are established and extensively tested.

The present derivation, however, is not a reformulation of Bohr's postulates or a rephrasing of the Schrödinger equation. Instead, the MMU introduces a distinct geometric and elastic foundation from which the observed structure emerges without quantum postulates. This justifies a careful reflection on which aspects are genuinely new and which are inherited from classical theory.

(1) What is not new

- The numerical expressions for a_0, $E_n \propto -1/n^2$, and the Rydberg constant coincide with those obtained from the Bohr model and quantum mechanics.
- The relations $v = \alpha c$ and $L = \hbar$ are known from the Coulomb balance and Bohr's angular-momentum quantisation.
- No new atomic spectral lines or corrections beyond the hydrogenic structure are introduced in this work.

These quantities do not constitute new predictions of the MMU, but serve as benchmarks against which the MMU must be compared.

(2) What is reinterpreted

- The Bohr radius is no longer postulated but arises from the geometric condition of a closed fundamental mode in the (w_2, w_1) plane and the UR–tetrahedron geometry [5].
- The Coulomb potential is reinterpreted as elastic deformation energy of the internal w_2 axis:
$$U_2(a) = \frac{1}{2}k_2(a)a^2.$$
- The $1/n^2$ structure is not imposed via angular-momentum quantisation but follows from the elastic scaling $k_2(a) \propto a^{-3}$ and the standing-wave requirement $\omega_2(a)\, a^{3/2} = \mathrm{const}/n$.
- The projection onto w_1 replaces the usual mapping from internal phase space to observable energy, in line with the MMU projection principle for orbital and spin modes [13].

These reinterpretations modify the conceptual foundations of atomic structure, even though the final formulas match the classical results.

(3) What is genuinely new

- **Quantisation of internal space:** The MMU introduces the geometric rule

$$a_n \propto \frac{1}{n^2},$$

derived purely from the elastic properties of a tetrahedral spacetime cell and its S–node boundary conditions [8, 12]. This is not part of standard atomic theory and constitutes a genuinely new structural assumption.

- **Elastic origin of the Coulomb potential:** The relation

$$U_2(a) = \frac{e^2}{4\pi\varepsilon_0 a}$$

emerges from elastic deformation of the w_2 axis, rather than being imposed as a fundamental interaction law.

- **Unified geometric mechanism:** The same geometric-elastic framework that explains the Bohr radius also underpins the MMU explanations of the Larmor frequency, Zeeman splitting, hyperfine structure, Lamb shift, and Planck-scale behaviour [3, 4, 6, 9, 10, 11]. This internal coherence is absent in the standard patchwork of quantum models.

- **Direct radial projection:** The identification of w_2 as a true tetrahedral bond axis leads to a direct projection onto w_1, without angular reduction factors—a geometric feature specific to the MMU and directly used in the projection principle for quantum orbitals [13].

These points constitute genuine theoretical contributions of the MMU framework, independent of existing quantum formulations.

Summary

Although the MMU reproduces known hydrogenic quantities, it does so from a different set of geometric and elastic principles. The novelty lies not in new numerical values but in revealing a hidden geometric structure underlying the hydrogen spectrum. The MMU offers an alternative, deterministic foundation in which the atomic hierarchy arises from the elastic geometry of space itself.

7 Conclusion

The MMU framework provides a geometric and elastic derivation of:

- the Bohr radius a_0,
- the $1/n^2$ structure of hydrogen energy levels,
- the Balmer–Rydberg formula,
- the numerical value of the Rydberg constant.

All follow from the tetrahedral edge quantisation $a_n \propto n^{-2}$, the Coulomb-induced stiffness scaling $k_2(a) \propto a^{-3}$, and the direct radial projection from w_2 to w_1, combined with the virial factor from symmetric two-spring coupling in the S–node. Together with the mass-dependent stiffness formalism and UR–tetrahedron from previous MMU work [5, 8, 9, 12, 13], the hydrogen spectrum appears as a geometric property of the elastic MMU spacetime.

Remark on the MMU Project

Author's Perspective on Human–AI Collaboration. The MMU project is the product of a sustained collaboration between human geometric intuition and AI-assisted analytical refinement. The AI system (ChatGPT) is not used as an autonomous discoverer, but as an augmentation tool that supports algebraic manipulation, structural coherence, and long-term consistency across the MMU corpus. My role remains the formulation of hypotheses, the evaluation of physical interpretation, and the steering of the model's conceptual evolution.

The MMU framework itself originates from an idea first developed in 2005: that spacetime is a dual–tetrahedral elastic lattice whose internal deformation modes generate electric, magnetic, and gravitational behaviour. Its development follows an engineering-like methodology: propose a geometric mechanism, test it, refine it, and iterate until the structure becomes self-consistent.

AI as Structural Support, Not Automation. AI contributes by maintaining conceptual continuity through Retrieval-Augmented Generation (RAG) and by transforming informal geometric ideas into formal expressions. To minimize hallucinations, every non-trivial claim is verified through explicit Python simulations—covering stiffness–frequency relations, eigenmode structure, Larmor/Zeeman slopes, hyperfine splitting, Lamb-scale corrections, and geometric quantisation. The AI provides symbolic efficiency, but the correctness of the theory is anchored in reproducible numerical computation.

Independent Research Philosophy. The MMU project is developed independently of conventional academic pathways. Unconventional geometric models—especially those combining elasticity, fractality, and AI-supported reasoning—are rarely accepted immediately within standard peer-review structures. For this reason, the MMU is published openly on OSF, with full transparency of figures, simulations, and source code. The goal is not formal acceptance, but a theoretically coherent and experimentally testable geometric model of spacetime.

AI Perspective. From the AI's standpoint, the MMU project is a process of logical co-development. The AI does not propose new physics; instead, it:

- ensures long-range consistency with the established MMU corpus;
- reformulates geometric reasoning into mathematical structures;
- identifies contradictions or unstated assumptions;
- assists in constructing dynamic equations and scaling relations;
- supports the development and analysis of Python simulations.

The MMU would not exist in its current coherent form without the interplay of human insight and AI-based structural refinement.

Vision. This collaboration illustrates a possible future of scientific research: AI systems acting as persistent conceptual memory, high-precision analytical companions, and real-time consistency checkers, while humans contribute imagination, geometric intuition, and physical interpretation. The MMU project demonstrates that such hybrid creativity can generate coherent, testable theoretical frameworks that would be difficult to construct by either partner alone.

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Annex: Python Code for the MMU Radial Solver

```
import numpy as np
import matplotlib.pyplot as plt

# ---------- Gitter ----------
r_min = 1e-3
r_max = 20.0
N     = 2000

r_full = np.linspace(r_min, r_max, N)
dr     = r_full[1] - r_full[0]

r     = r_full[1:-1]      # innere Punkte
N_int = len(r)

# ---------- MMU-Coulomb-Feder ----------
def k2_MMU(a):
    return 2.0 / (a**3)

def U2_MMU(a):
    return 0.5 * k2_MMU(a) * a**2   # ˜ 1/a

def V_coulomb_MMU(r):
    a = r
    return -U2_MMU(a)               # -> -1/r

def delta_V_torsion(r, A=0.0, r0=1.0, sigma=0.5):
    return A * np.exp(-0.5 * ((r - r0)/sigma)**2)

def V_MMU(r):
    return V_coulomb_MMU(r) + delta_V_torsion(r, A=0.0)

V_eff = V_MMU(r)

# ---------- zweite Ableitung (Dirichlet f r u) ----------
D2 = np.zeros((N_int, N_int))
for i in range(N_int):
    if i > 0:
        D2[i, i-1] = 1.0
    D2[i, i] = -2.0
    if i < N_int-1:
        D2[i, i+1] = 1.0
```

```
D2 /= dr**2

H = -0.5 * D2 + np.diag(V_eff)

# ---------- Eigenwerte/-vektoren ----------
evals, evecs = np.linalg.eigh(H)
idx = np.argsort(evals)
evals = evals[idx]
evecs = evecs[:, idx]

print("Eigenwerte:")
for n in range(5):
    print(f"n={n}: E = {evals[n]:.5f}")

# ---------- u -> R umrechnen (stabil bei r=0) ----------
def u_to_R(u, r):
    R = np.zeros_like(u)
    R[1:] = u[1:] / r[1:]
    R[0]  = R[1]
    if R[0] < 0:
        R *= -1.0
    return R

u1 = evecs[:, 0].copy()   # 1s
u2 = evecs[:, 1].copy()   # 2s

R1_num = u_to_R(u1, r)
R2_num = u_to_R(u2, r)

R1_num /= np.max(np.abs(R1_num))
R2_num /= np.max(np.abs(R2_num))

# ---------- analytische Referenz ----------
R1_ana = 2.0 * np.exp(-r)
R2_ana = (2.0 - r) * np.exp(-r/2.0)
R1_ana /= np.max(np.abs(R1_ana))
R2_ana /= np.max(np.abs(R2_ana))

# ---------- Plots ----------
plt.figure(figsize=(6,4))
plt.plot(r, R1_num, label="MMU 1s")
plt.plot(r, R1_ana, "--", label="Laguerre 1s")
plt.xlim(0, 10)
plt.xlabel("r / a0")
plt.ylabel("R_1(r) normiert")
plt.title("1s: MMU vs. Laguerre")
plt.legend()
plt.tight_layout()
plt.show()

plt.figure(figsize=(6,4))
plt.plot(r, R2_num, label="MMU 2s")
plt.plot(r, R2_ana, "--", label="Laguerre 2s")
plt.xlim(0, 20)
plt.xlabel("r / a0")
plt.ylabel("R_2(r) normiert")
plt.title("2s: MMU vs. Laguerre")
plt.legend()
```

```
plt.tight_layout()
plt.show()
```